Name-Bhakare Purva Shivaji

Roll no-08

Practical no-10

```
import os
import zipfile
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
# 1. Load dataset from ZIP file
zip_path = r"C:\Users\PVGIT\Downloads\archive (18).zip"
extract_path = r"C:\Users\PVGIT\Downloads\pca_data"
with zipfile.ZipFile(zip_path, 'r') as z:
    z.extractall(extract_path)
# Find the first CSV file
csv_file = next(
    os.path.join(root, f)
    for root, _, files in os.walk(extract_path)
    for f in files if f.endswith('.csv')
)
df = pd.read_csv(csv_file)
print("Dataset shape:", df.shape)
print(df.head())
# 2. Separate features and target (last column)
X = df.iloc[:, :-1].copy()
y = df.iloc[:, -1].copy()
# Convert categorical features to numeric
X = pd.get_dummies(X, dummy_na=True)
X = X.fillna(X.median())
```

```
# Convert target labels into numeric value y =
LabelEncoder().fit_transform(y.fillna("Unknown").astype(str))

# Standardize features

X_scaled = StandardScaler().fit_transform(X)

print("\nFeatures BEFORE PCA:", X_scaled.shape[1])

# 3. Apply PCA (retain 95% variance)

pca = PCA(n_components=0.95)

X_pca = pca.fit_transform(X_scaled)

print("Features AFTER PCA:", X_pca.shape[1])

print("Explained variance ratio:",

     pca.explained_variance_ratio_)

print("Total variance retained:",

     round(sum(pca.explained_variance_ratio_) * 100, 2), "%")

# 4. Plot explained variance

plt.figure(figsize=(7, 4))

plt.plot(

   range(1, len(pca.explained_variance_ratio_) + 1),

   pca.explained_variance_ratio_,

   marker='o'

)

plt.xlabel("Principal Component")

plt.ylabel("Explained Variance Ratio")

plt.title("PCA Explained Variance")

plt.grid()

plt.tight_layout()

plt.show()

# 5. Visualize dataset in 2D using PCA

pca_2d = PCA(n_components=2)

X_2d = pca_2d.fit_transform(X_scaled)

plt.figure(figsize=(7, 5))

scatter = plt.scatter(
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    X_2d[:, 0], X_2d[:, 1],
    c=y, cmap='viridis', alpha=0.7
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Dataset Visualization Before/After PCA")
plt.colorbar(scatter, label="Class")
plt.tight_layout()
plt.show()
```

Output-

```
Dataset shape: (270, 14)
   Age  Sex  Chest pain type   BP  Cholesterol  FBS over 120  EKG results  \
0   70    1               4  130          322             0             2
1   67    0               3  115          564             0             2
2   57    1               2  124          261             0             0
3   64    1               4  128          263             0             0
4   74    0               2  120          269             0             2

   Max HR  Exercise angina  ST depression  Slope of ST  \
0     109               0            2.4            2
1     160               0            1.6            2
2     141               0            0.3            1
3     105               1            0.2            2
4     121               1            0.2            1

   Number of vessels fluro  Thallium Heart Disease
0                        3       3      Presence
1                        0       7       Absence
2                        0       7      Presence
3                        1       7       Absence
```

4 1 3 Absence

Features BEFORE PCA: 13

Features AFTER PCA: 12

Explained variance ratio: [0.23425531 0.12366676 0.095472 0.08928379 0.0778754 0.06989303 0.0635299 0.05764219 0.0521241 0.04431827 0.03488548 0.03098712]

Total variance retained: 97.39 %



